

EV-MESH: Autonomous Mobile Energy Orchestration Hub

Problem Statement ID – [Your ID] | Team ID – [Your Team ID] | Team Name – [Your Team Name]

THE PROBLEM



Stranded EVs
No nearby charging access



Wasted Solar
Unused rooftop generation



Grid Overload
Uncoordinated charging spikes

THE SOLUTION

- ☐ An intelligent energy-routing system that transforms EVs into mobile energy-storage nodes

MAREN ENGINE - The Core Innovation

Mobility-Aware Reserve Energy Negotiation

An intelligent algorithm that calculates safe energy transfers by analyzing:

- Current battery state of charge & required safety reserve
- Planned departure time & estimated travel distance
- Battery health & converter safety limits

Result: Transfers only minimum safe surplus, ensuring donor vehicles are never stranded while enabling emergency assistance.

HOW IT WORKS

01

Battery-aware routing

Calculates current battery state & safety reserve

02

Trip-aware planning

Considers planned departure time & travel needs

03

Stranding prevention

Ensures donor vehicles are never stranded

KEY RESEARCH AREAS

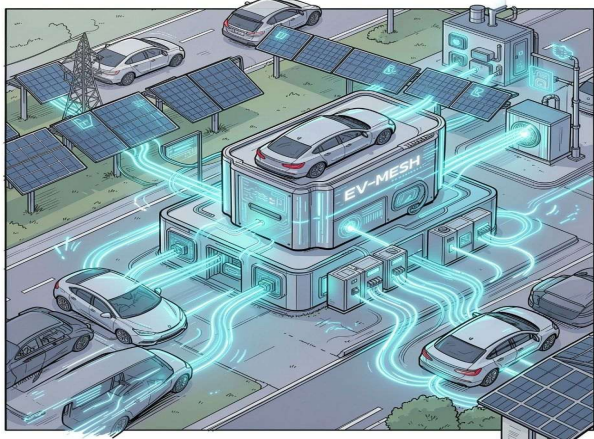
V2G & Smart Grid Integration

Bidirectional Power Topologies

AI & Control Algorithms

TECHNICAL APPROACH

- Edge-embedded controllers
ESP32 + Raspberry Pi
- Bidirectional converters
Isolated DC-DC power flow
- Safety monitoring
Real-time monitoring & auto-cutoff



IMPACT

Social

Emergency energy sharing, democratized EV ownership

Economic

15-20% grid peak reduction, ₹2-3/kWh savings

Environmental

25-30% renewable utilization increase